

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science and Engineering

Detailed Project Report Structure

and Visual Presentation Plan

Machine Learning-Based Daily Traffic and Toll Revenue Forecasting for the
Padma Bridge

Course: CSE 4112: Machine Learning Laboratory

Group: B2

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1 Introduction

Include the following:

- Importance of Padma Bridge forecasting.
- Why traffic changes during weekends, holidays and Eid.
- Why railway, Jamuna traffic and regional economic activity are examined.
- A short overview of the complete project.

Recommended length: 2 pages.

2 Problem Statement and Objectives

Include the following:

- Main research questions.
- Forecasting objectives.
- Event-aware objective.
- Transport and regional-analysis objectives.

Recommended length: 1–2 pages.

3 Related Work

Briefly discuss previous applications of:

- Traffic forecasting.
- Random Forest and XGBoost.
- Time-series forecasting.
- Event-aware forecasting.
- Infrastructure impact analysis.
- Nighttime lights as an economic proxy.

Recommended length: 2–3 pages.

4 Dataset Description

Include the following:

- Padma Bridge dataset.
- Jamuna Bridge dataset.
- Weather and holiday data.
- Railway intervention date.
- Nighttime-light dataset.
- Dataset sizes and date coverage.
- Target variables.

Use one dataset summary table.

Recommended length: 2–3 pages.

5 System Architecture

Include the following:

- One complete architecture diagram.
- Data-input layer.
- Preprocessing layer.
- Feature-engineering layer.
- Forecasting models.
- Event-aware models.
- Statistical-analysis branches.
- Evaluation and outputs.

Recommended length: 2 pages.

6 Data Preprocessing and Feature Engineering

6.1 Numeric Cleaning

Explain how traffic and toll strings are converted into numerical values.

6.2 Date Cleaning and Alignment

Explain:

- Date-format conversion.
- Sorting.
- Missing-date detection.
- Duplicate checking.
- Continuous calendar creation.

6.3 Data-Quality Audit

Explain the total-versus-side consistency checks.

6.4 Dataset Integration

Explain how Padma, Jamuna, weather and event data are merged.

6.5 Feature Engineering

Explain feature groups rather than listing all 51 features:

- Calendar and cyclical features.

- Weather features.
- Eid and holiday features.
- Lag features.
- Rolling features.
- Jamuna history.
- Railway features.
- Revenue-per-vehicle features.

6.6 Data-Leakage Prevention

Clearly state that historical variables are shifted by at least one day.

Recommended length: 3–4 pages.

7 Methodology

7.1 Forecasting Formulation

Define the inputs, next-day traffic output and next-day toll output.

7.2 Forecasting Models

Briefly explain:

- Persistence.
- Seasonal baseline.
- Linear Regression.
- SARIMAX.
- Random Forest.
- XGBoost.
- Residual MLP.

7.3 Event-Aware Forecasting

Explain:

- Ordinary XGBoost.
- Weighted XGBoost.
- Ordinary CatBoost.
- Weighted CatBoost.
- Event weights.

7.4 Weather and Event Analysis

Explain group comparison, Spearman correlation, SHAP and the Eid event window.

7.5 Padma–Jamuna Analysis

Explain ADF, cross-correlation, Granger causality, VAR and impulse response.

7.6 Railway Analysis

Explain Interrupted Time Series and HAC standard errors.

7.7 Nighttime-Light Analysis

Explain Difference-in-Differences, fixed effects and pre-trend testing.

Recommended length: 5–6 pages.

8 Experimental Setup

Include the following:

- Training, validation and test dates.
- Chronological split.
- Time-series cross-validation.
- Hyperparameter tuning.
- Hardware and software.
- Evaluation metrics.

Only show the final important configurations. Put full hyperparameters in the appendix.

Recommended length: 2 pages.

9 Results and Discussion

This should be the main combined section.

9.1 Traffic Forecasting Results

Use this order:

1. Present the comparison table.
2. Show the actual-versus-predicted traffic graph.
3. Explain the result.
4. Identify the best model.

Example discussion:

Random Forest produced the lowest traffic MAE and MAPE, with an average error of 1,451 vehicles and a MAPE of 8.58%. XGBoost produced the lowest RMSE and highest R^2 , indicating that it handled large traffic variations more effectively.

9.2 Toll Forecasting Results

Include the model-comparison table, actual-versus-predicted toll graph and best-model explanation. Main result:

Residual MLP produced the lowest toll MAE and MAPE, while XGBoost produced the strongest RMSE and R^2 .

9.3 Feature-Ablation and Explainability Results

Include:

- Feature-ablation table.
- A compact table of the most important traffic features.

Discuss that weather alone was insufficient, calendar features improved performance, Padma history gave the largest improvement, Jamuna history provided additional information and toll history improved toll prediction.

9.4 Weather, Weekend, Holiday and Eid Results

Include one result table and the Eid event-profile graph. Discuss that traffic rises before Eid, the peak occurs approximately two days before Eid, weather relationships are weaker than calendar effects, and these are associations rather than causal effects.

9.5 Event-Aware Forecasting Results

Include:

- Overall model-comparison table.
- A combined traffic-and-toll MAPE graph by day type.

Discuss that Weighted XGBoost improved Eid forecasting, performed strongly for toll revenue, did not improve general public-holiday traffic consistently, and weighting should be selected separately for each target. Also mention the difference between Alvir’s original XGBoost and Siyam’s retrained XGBoost results and reconcile them before declaring a final traffic winner.

9.6 Padma–Jamuna Relationship Results

Include a compact statistical summary table. Discuss the strong same-day relationship, significant Granger relationships in both directions, the 14-lag VAR and the fact that predictive relationship does not imply physical causation.

9.7 Railway Impact Results

Include the ITS coefficient table and railway-intervention graph. Discuss that the immediate change was not significant, the post-rail trend change was significantly negative, and the result is an association rather than proven causation.

9.8 Nighttime-Light Results

Include the before-and-after values, DiD coefficient, confidence interval, pre-trend result and the southwest-versus-control trend graph. Discuss that both groups became brighter, the southwest

increased less, parallel trends were not satisfied, and the result cannot support a strong causal conclusion.

9.9 Integrated Discussion

Use approximately one page to connect the following findings:

- Forecasting results are strong.
- Historical traffic is the most important input.
- Eid and calendar factors are more important than weather.
- Event weighting helps selectively.
- Padma and Jamuna have predictive dependence.
- Railway and NTL results require cautious interpretation.
- Prediction findings are stronger than causal-impact findings.

Recommended length for the complete section: 10–12 pages.

10 System Output and Demonstration

Show one clear example of system operation.

Input

- Prediction date.
- Expected weather.
- Holiday/Eid status.
- Historical values loaded automatically.

Output

- Predicted traffic.
- Predicted toll revenue.
- Model used.
- Day/event type.

Use a dashboard-style output figure or table.

Recommended length: 1–2 pages.

11 Limitations and Future Work

Include:

- Limited number of Eid events.
- Only four years of traffic data.
- Inconsistent source records.
- Weather-forecast uncertainty.

- No rail-passenger data.
- No control corridor for railway analysis.
- Granger causality is not physical causation.
- Failed NTL parallel-trend assumption.
- No prediction intervals.
- Only one-day forecasting currently supported.

Future work:

- Multi-horizon forecasting.
- Directional forecasting.
- Prediction intervals.
- Improved non-Eid holiday modelling.
- Vehicle-category data.
- Periodic retraining.

Recommended length: 2 pages.

12 Conclusion

Summarize the inputs used, models compared, best forecasting models, main event-aware improvement, main relationships found and practical usefulness.

Recommended length: 1 page.

13 References

Recommended length: 2–3 pages.

14 Suggested Page Allocation

15 Structure for Every Result Subsection

Use the same pattern every time:

1. **Question:** What are we trying to discover?
2. **Method reminder:** Which model or technique was used? Keep this to two or three sentences because the complete method was explained earlier.
3. **Numerical result:** Present one concise table.
4. **Visual result:** Present one relevant graph when necessary.
5. **Interpretation:** Explain what the numbers and graph mean.
6. **Limitation:** State what cannot be concluded.

Example:

Railway Impact Results

Question: Did road traffic change after railway service began?

Table 1: Suggested allocation for a report under 40 pages

Section	Approximate pages
Title, abstract and contents	3–4
Introduction	2
Problem and objectives	1–2
Related work	2–3
Datasets	2–3
System architecture	2
Preprocessing and features	3–4
Methodology	5–6
Experimental setup	2
Results and discussion	10–12
System demonstration	1–2
Limitations and future work	2
Conclusion	1
References	2–3
Approximate total	35–39

Method: Interrupted Time Series with weather, calendar and Eid controls.

Result: Immediate change = -3.37% , $p = 0.460$; trend change = -0.0367% per day, $p = 0.0052$.

Graph: Monthly traffic with the railway-opening line.

Interpretation: There was no significant immediate decrease, but there was a significant negative change in the later traffic trend.

Limitation: Without rail-passenger data or a control corridor, this is an association rather than proven causation.

Results and their discussion should be combined under one main Results and Discussion section, with a separate subsection for each component. Finish with a short integrated discussion connecting all findings.

16 Figure Names, Availability and Locations

Figure 1: Unified system architecture and preprocessing workflow

Availability: Not currently present.

Location or action: Create one new image combining data sources, cleaning, date alignment, feature engineering, chronological splitting, forecasting models, analytical branches, evaluation and outputs.

Figure 2: Actual versus predicted traffic and toll revenue

Availability: Present as two separate existing images.

Location or action: Use the two existing files from Alvir's project ZIP:

outputs/figures/traffic_actual_vs_predicted.png

outputs/figures/toll_actual_vs_predicted.png

Combine them into one two-panel figure.

Figure 3: Padma Bridge traffic around Eid

Availability: Present.

Location or action: Use outputs/figures/eid_event_profile.png from Alvir's project ZIP.

Figure 4: Event-aware traffic and toll MAPE by day type

Availability: Present as two separate figures in Siyam's report.

Location or action: Combine Figure 1 (traffic MAPE by segment) and Figure 3 (toll MAPE by segment) from Siyam_Event_Aware_Forecasting_Report.pdf into one two-panel figure.

Figure 5: Padma road traffic around railway-service introduction

Availability: Present.

Location or action: Use outputs/figures/railway_intervention_monthly_traffic.png from Alvir's project ZIP.

Figure 6: Nighttime-light trends for southwestern and control districts

Availability: Present.

Location or action: Use outputs/figures/ntl_treatment_control_trends.png from Alvir's project ZIP.

17 Tables to Include in the Final Report

Table 2: Dataset summary

Required content: Dataset name, row count, date coverage and purpose for Padma, Jamuna and nighttime-light data.

Source or action: Use the dataset audit and processed-data information from Alvir’s project ZIP.

Table 3: Feature-group summary

Required content: The 22 calendar/event/weather/railway features, 29 historical Padma/Jamuna features and total of 51 model inputs.

Source or action: Summarize the feature definitions in `src/config.py`; place the full feature list in the appendix.

Table 4: Chronological data split

Required content: Training, validation and test rows, date ranges and purpose.

Source or action: Use Alvir’s forecasting run details: 997 training, 214 validation and 214 test rows.

Table 5: Traffic forecasting model comparison

Required content: Model, MAE, RMSE, MAPE and R^2 for all forecasting models.

Source or action: Use `outputs/tables/traffic_test_metrics.csv` from Alvir’s project ZIP.

Table 6: Toll-revenue forecasting model comparison

Required content: Model, MAE, RMSE, MAPE and R^2 for all forecasting models.

Source or action: Use `outputs/tables/toll_test_metrics.csv` from Alvir’s project ZIP.

Table 7: Feature-ablation results

Required content: Traffic and toll performance for weather only, weather plus calendar, Padma history, Jamuna history and toll history.

Source or action: Combine `traffic_feature_ablation.csv` and `toll_feature_ablation.csv` from `outputs/tables/`.

Table 8: Weather and event relationships

Required content: Traffic differences for weekends, public holidays, Eid-labelled days and heavy-rain days.

Source or action: Use `outputs/tables/weather_holiday_group_comparison.csv` from Alvir’s project ZIP.

Table 9: Event-aware forecasting comparison

Required content: Overall, normal-day, non-Eid holiday and Eid-window MAPE for Ordinary XGBoost, Weighted XGBoost, Ordinary CatBoost, Weighted CatBoost and the appropriate existing reference model.

Source or action: Use Siyam’s report tables, but reconcile the original and retrained XGBoost traffic results before declaring the final winner.

Table 10: Statistical-analysis summary

Required content: Technique, inputs, principal coefficient or test result, p-value and interpretation for Padma–Jamuna, railway ITS and nighttime-light DiD analyses.

Source or action: Use the Granger/VAR, railway ITS and NTL DiD output files in Alvir’s project ZIP.